

deepened_round2

Deepened inventory (round 2)

Round-2 deepening of packages flagged shallow/partial/unknown in round 1 (internal_services.md coverage notes + the owned-submodule umbrella rows). Each previously-thin entry is enumerated into discrete features in the SAME 12-column schema as internal_services.md (legend in docs/features/Status.md § “Status dimensions”). Assessed from real code evidence (impl reality, wiring, *_test.go presence) per CONST-035 / §11.4.107 anti-bluff.

Anti-bluff caveats (read before trusting a row). - 📹 Video is **no** for every row — no analyzed recording exists; the conductor owns video confirmation. Overall is therefore **never confirmed**; honest rollups are working-untaped / partial / gap. - Origin is native (all own-org code). - **Round-1 stub claims corrected here from direct source reads** (a round-2 exploration subagent reported these as stubs WITHOUT reading the code — verified false): workspace docker Stop/Remove are real (docker stop / docker rm -f, manager.go:21–37); voice recorder Stop is real (Process.Signal(os.Interrupt) then Kill(), recorder.go:63–80); roocode CodeReviewer.Review is real (reads the file + scans for TODO/issue patterns, reviewer.go:16–30). They are done/partial (impl real) but Real-use=unknown (shipped-flow wiring unconfirmed), not stubs. - Submodule-side Wired follows internal_services.md’s wiring model: helix_agent is imported directly (Wired=yes where HelixCode consumes it), transitively-reachable own-org deps are Wired=partial, repo-present-but-not- in-build-graph are Wired=no. helix_specifier/helix_qa are imported by module path (Wired=yes); panoptic is a Challenge/recording submodule not in HelixCode’s Go build graph (Wired=no). - Submodule Tests reflects *_test.go presence in/near the package; it is NOT a coverage-percentage claim. README-cited coverage figures (helix_agent “65.6%”, panoptic “78%”) are NOT used as evidence — unverified.

Internal packages (round-1 partial/unknown → deepened)

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&S
infrastructure	internal/clientcore	agentic tool registry wiring (git/fs/LSP/MCP, live-built)	done	yes	yes	unit	no
infrastructure	internal/clientcore	LSP read-only diagnostics tool wiring	done	yes	unknown	unit	no
infrastructure			done	yes	unknown	unit	no

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V
	internal/ clientcore	MCP config-merge + server startup + tool registration					
infrastructure	internal/ clientcore	tool-loop system-prompt generation (from registry names)	done	yes	yes	unit	no
infrastructure	internal/ clientcore	tool-trace adapter (agent → ensembleui)	done	yes	yes	unit	no
infrastructure	internal/ clientcore	skills + plugins loader (graceful-on-failure)	done	yes	unknown	unit	no
infrastructure	internal/ clientcore	verifier adapter wiring (CONST-036/037 single-source)	done	yes	unknown	unit	no
infrastructure	internal/ clientcore	env-provider registration (cloud API keys, zero-hardcode)	done	yes	unknown	unit	no
infrastructure	internal/ clientcore	HelixAgent provider registration (when server reachable)	done	partial	unknown	unit	no
infrastructure	internal/ agentbridge	verifier bridge config wrapper	done	yes	unknown	unit	no
infrastructure	internal/ agentbridge	VerifyModel real HTTP call to verifier service	done	yes	unknown	unit	no
infrastructure	internal/ agentbridge	test request/result marshal+unmarshal	done	yes	unknown	unit	no
infrastructure	internal/ checkpoint	git-backed snapshot (stash/write-tree/read- tree)	done	yes	yes	unit	no
infrastructure	internal/ checkpoint	file-copy fallback backend (real os.Read/ Write)	done	yes	yes	unit	no
infrastructure	internal/ checkpoint	checkpoint create (capture)	done	yes	yes	unit	no
infrastructure	internal/ checkpoint	checkpoint restore (undo, writes real bytes)	done	yes	yes	unit	no
infrastructure	internal/ checkpoint	metadata persistence (label/timestamp JSON)	done	yes	yes	unit	no
infrastructure	internal/ checkpoint	multi-backend auto- selection (git else files)	done	yes	yes	unit	no
service	internal/ ensembleui	tool-trace formatting (text/markdown render)	done	yes	unknown	unit	no
service	internal/ ensembleui	metadata extraction helpers (int/string/slice)	done	yes	unknown	unit	no
service	internal/ ensembleui	multi-model response assembly	done	yes	unknown	unit	no

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V
service	internal/ensembleui	per-model token-usage aggregation	done	yes	unknown	unit	no
infrastructure	internal/substrate	Unit interface (task contract)	done	yes	unknown	unit	no
infrastructure	internal/substrate	queue-based scheduler (enqueue + dispatch)	done	yes	unknown	unit	no
infrastructure	internal/substrate	serial FIFO execution (single worker)	done	yes	unknown	unit	no
infrastructure	internal/substrate	context-cancellation propagation to Execute	done	yes	unknown	unit	no
infrastructure	internal/workspace	workspace create (project bootstrap)	done	partial	unknown	unit	no
infrastructure	internal/workspace	docker runner start/stop/remove/list (real os/exec)	done	partial	unknown	unit	no
infrastructure	internal/workspace	workspace status tracking (enum + DB persist)	done	partial	unknown	unit	no
infrastructure	internal/workspace	CLI tools (create/delete/list registration)	done	yes	unknown	unit	no
infrastructure	internal/workspace	i18n translator seam	done	yes	yes	unit	no
infrastructure	internal/workspace	auto-cleanup / TTL teardown	partial	no	unknown	unit	no
service	internal/voice	audio device detection (arecord/sox/parec probe)	done	partial	unknown	unit	no
service	internal/voice	audio capture start (real exec.Command)	done	partial	unknown	unit	no
service	internal/voice	audio capture stop (Signal(Interrupt)→Kill)	done	partial	unknown	unit	no
service	internal/voice	Whisper API transcription (real multipart HTTP POST)	done	partial	unknown	unit	no
service	internal/voice	WAV validation before transcribe	done	partial	unknown	unit	no
service	internal/voice	local whisper.cpp fallback transcription	partial	no	unknown	unit	no
service	internal/roocode	in-memory conversation store (create/add-message)	done	partial	unknown	unit	no
service	internal/roocode	task delegation (TaskSpec build; no real dispatch)	partial	no	unknown	unit	no
service			partial	no	unknown	unit	no

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V
	internal/ roocode	code generation (Generate/Bootstrap scaffold)					
service	internal/ roocode	code review (file read + TODO/issue scan)	done	partial	unknown	unit	no
service	internal/ roocode	i18n translator seam	done	yes	yes	unit	no
service	internal/ telemetry	OpenTelemetry provider bootstrap (OTLP exporter)	done	yes	unknown	unit	no
service	internal/ telemetry	agent instrumentation (spans + iteration counter)	done	yes	unknown	unit	no
service	internal/ telemetry	LLM instrumentation (traced-provider wrapper)	done	yes	unknown	unit	no
service	internal/ telemetry	tool instrumentation (call spans)	done	yes	unknown	unit	no
service	internal/ telemetry	config-from-env (OTEL_* vars + exporter-kind resolve)	done	yes	yes	unit	no
infrastructure	internal/ verifier	flexible timestamp parse (Unix + RFC3339)	done	yes	yes	unit	no
infrastructure	internal/ verifier	adapter facade (cache+health+client, IsEnabled/Reachable)	done	yes	unknown	unit	no
infrastructure	internal/ verifier	two-tier cache (TTL + Redis)	done	partial	unknown	unit	no
infrastructure	internal/ verifier	health monitor (background liveness goroutine)	done	partial	unknown	unit	no
infrastructure	internal/ verifier	background poller (interval model-sync)	done	partial	unknown	unit	no
infrastructure	internal/ verifier	bootstrap (embedded-or- remote stack init)	done	yes	unknown	unit,integ	no
infrastructure	internal/ verifier	embedded verifier server (fallback)	partial	partial	unknown	unit	no
infrastructure	internal/ verifier	fallback model list (offline mode)	done	yes	yes	unit	no
infrastructure	internal/ verifier	real-server integration test (live LLMsVerifier)	done	yes	unknown	integ	no
service	internal/ worker	consensus manager (Raft state machine, election/heartbeat)	done	partial	unknown	unit	no
service			done	partial	unknown	unit	no

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V
	internal/ worker	vote-transport abstraction (decoupled from wire)					
service	internal/ worker	single-node fallback (nil-transport safe step- down)	done	yes	yes	unit	no
service	internal/ worker	worker isolation: sandbox dir create (0750)	done	partial	unknown	unit	no
service	internal/ worker	worker isolation: resource limits (cgroup writes)	done	partial	unknown	unit	no
service	internal/ worker	SSH client pool (persistent connection reuse)	done	yes	yes	unit	no
service	internal/ worker	consensus stress + chaos tests (concurrent votes, SIGKILL)	done	yes	yes	unit	no
service	internal/ worker	worker-pool isolation stress test	done	yes	yes	unit	no
service	internal/ server	Gin server + JWT auth middleware	done	yes	unknown	unit,integ	no
service	internal/ server	project list/register handlers (user-from- context)	done	yes	yes	integ	no
service	internal/ server	LLM generate endpoint (real provider stream)	done	yes	yes	unit,integ	no
service	internal/ server	LLM model-list endpoint (verifier- queried)	done	yes	yes	unit,integ	no
service	internal/ server	worker/task/project CRUD handlers	done	yes	yes	integ	no
service	internal/ server	QA session start handler (helix_qa integration)	done	yes	unknown	unit	no
service	internal/ server	specify endpoint (spec from prompt)	done	yes	yes	unit,integ	no
service	internal/ server	stats / pprof / error- categorize (400-vs-500) helpers	done	yes	yes	unit	no
service	internal/ server	i18n translator seam (context-aware)	done	yes	yes	unit	no

Owned submodules (round-1 umbrella rows → deepened principal packages)

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V
submodule	helix_agent	internal/llm multi-provider interface (Claude/DeepSeek/Gemini/Mistral/Qwen/xAI/OpenRouter/Ollama/llama.cpp/Bedrock/Azure)	done	yes	unknown	unit,integ	no
submodule	helix_agent	internal/ensemble multi-model debate/fusion orchestrator	done	yes	yes	unit,integ,e2e	no
submodule	helix_agent	internal/provider+providers per-provider abstraction layer	done	yes	unknown	unit	no
submodule	helix_agent	internal/verifier LLMsVerifier single-source-of-truth	done	yes	unknown	unit	no
submodule	helix_agent	internal/server HTTP(Gin) +WebSocket API (completions/ensemble/stream)	done	yes	unknown	unit,integ,e2e	no
submodule	helix_agent	internal/handlers per-route request/response handlers	done	yes	unknown	unit	no
submodule	helix_agent	internal/database PostgreSQL persistence (pgx/v5, migrations)	done	partial	unknown	integ	no
submodule	helix_agent	internal/redis caching layer (go-redis/v9)	done	partial	unknown	integ	no
submodule	helix_agent	internal/auth JWT + bcrypt/argon2 + OAuth2	done	partial	unknown	unit,integ	no
submodule	helix_agent	internal/security PII redaction + guardrails bridge	done	partial	unknown	unit	no
submodule	helix_agent	internal/streaming SSE/WebSocket token-level streaming	done	yes	unknown	unit	no

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V
submodule	helix_agent	internal/cache semantic (embeddings dedup) caching	done	partial	unknown	unit	no
submodule	helix_agent	internal/knowledge RAG / vector-DB (zep) bridge	done	partial	unknown	unit	no
submodule	helix_agent	internal/monitoring Prometheus metrics	done	partial	unknown	unit	no
submodule	helix_agent	internal/logging structured JSON + gRPC tracing	done	partial	unknown	unit	no
submodule	helix_agent	internal/mcp Model Context Protocol server/client	done	partial	unknown	unit	no
submodule	helix_agent	internal/skills+plugins hot-reload plugin system	done	partial	unknown	unit	no
submodule	helix_agent	internal/analytics usage telemetry + warehouse export	partial	partial	unknown	unit	no
submodule	helix_agent	pkg/sdk Go + Python client SDKs	done	partial	unknown	unit	no
submodule	security	pkg/guardrails content guardrail engine (severity rules)	done	partial	unknown	unit	no
submodule	security	pkg/pii PII detect+redact (email/phone/SSN/CC-Luhn/IPv4)	done	partial	unknown	unit	no
submodule	security	pkg/content composable filter chains (length/pattern/keyword)	done	partial	unknown	unit	no
submodule	security	pkg/policy rule-based policy enforce (allow/deny/audit)	done	partial	unknown	unit	no
submodule	security	pkg/scanner vuln-scan interface + report aggregation	done	partial	unknown	unit	no

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V
submodule	security	pkg/headers HTTP security headers middleware (CSP/HSTS/...)	done	partial	unknown	unit	no
submodule	security	pkg/securestorage AES-256-GCM + Argon2id key derivation	done	partial	unknown	unit	no
submodule	security	pkg/e2ee end-to-end encryption (ChaCha20-Poly1305)	done	partial	unknown	unit	no
submodule	security	pkg/ssrf SSRF prevention (outbound HTTP guard)	done	partial	unknown	unit	no
submodule	security	pkg/attestation + pkg/gpuattest platform/GPU attestation	partial	no	unknown	unit	no
submodule	helix_specifier	parallel execution (bounded-concurrency dispatch)	done	yes	unknown	unit	no
submodule	helix_specifier	constitution-as-code (machine-readable rule enforce)	done	yes	unknown	unit	no
submodule	helix_specifier	Nyquist TDD ($\geq 2x$ test-to-impl ratio enforce)	done	yes	unknown	unit	no
submodule	helix_specifier	debate architecture (multi-round spec refinement)	done	yes	yes	unit	no
submodule	helix_specifier	skill learning (running-avg proficiency tracking)	done	yes	unknown	unit	no
submodule	helix_specifier	brownfield analysis (legacy pattern/dep detection)	done	yes	unknown	unit	no
submodule	helix_specifier	predictive specification (future-req prediction)	done	yes	unknown	unit	no
submodule	helix_specifier		done	yes	unknown	unit	no

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V
		cross-project transfer (shared knowledge base)					
submodule	helix_specifier	adaptive ceremony (dynamic level by quality metrics)	done	yes	unknown	unit	no
submodule	helix_specifier	spec memory (persistent index + semantic search)	done	yes	unknown	unit	no
submodule	helix_qa	pkg/autonomous autonomous QA session engine (~9.5k loc, 21 tests)	done	yes	unknown	unit	no
submodule	helix_qa	pkg/orchestrator QA run orchestration	done	yes	unknown	unit	no
submodule	helix_qa	pkg/navigator UI navigation/driving (~3.8k loc, 16 tests)	done	yes	unknown	unit	no
submodule	helix_qa	pkg/detector + pkg/issuedetector issue/error detection	done	yes	unknown	unit	no
submodule	helix_qa	pkg/recordingqa recording-validator (anti-bluff §11.4.107)	done	yes	unknown	unit	no
submodule	helix_qa	pkg/regression standing regression-guard suite	done	yes	unknown	unit	no
submodule	helix_qa	pkg/evidence captured-evidence management	done	yes	unknown	unit	no
submodule	helix_qa	pkg/challengegen Challenge generation	partial	yes	unknown	unit	no
submodule	helix_qa	pkg/reporter QA reporting	done	yes	unknown	unit	no
submodule	helix_qa	pkg/conduit + pkg/bridge sync-channel / agent bridge (§11.4.116)	done	yes	unknown	unit	no
submodule	helix_qa	pkg/replay + pkg/reproduce replay /	done	partial	unknown	unit	no

Area	Component	Feature	Dev	Wired	Real-use	Tests	V&V
		defect reproduction					
submodule	helix_qa	pkg/distributed + pkg/maestro distributed/maestro coordination	partial	partial	unknown	unit	no
submodule	helix_qa	banks/ test-bank corpus (JSON/ YAML suites)	done	yes	unknown	none	no
submodule	helix_qa	cmd/* tooling (axtree/capture/ recvalidate/ omniparser/uitars/ lpips/...)	done	partial	unknown	unit	no
submodule	panoptic	internal/launcher multi-platform test launcher (web/ desktop/mobile)	done	no	unknown	unit	no
submodule	panoptic	internal/executor UI automation executor	done	no	unknown	unit	no
submodule	panoptic	internal/vision computer-vision element detection	done	no	unknown	unit	no
submodule	panoptic	internal/ocr OCR text extraction	done	no	unknown	unit	no
submodule	panoptic	internal/recvalidate recording validator	done	no	unknown	unit	no
submodule	panoptic	internal/ai AI test generation/ enhancement	done	no	unknown	unit	no
submodule	panoptic	internal/cloud cloud storage/ integration	done	no	unknown	unit	no
submodule	panoptic	internal/platforms per-platform automation adapters	done	no	unknown	unit	no
submodule	panoptic	screenshot + video recording capture	done	no	unknown	unit	no
submodule	panoptic	cmd/record + cmd/ vision + cmd/ testgen CLI tooling	done	no	unknown	unit	no

Round-2 coverage notes

- **Internal deepening:** the 12 round-1-flagged packages (`clientcore`, `agentbridge`, `checkpoint`, `ensembleui`, `substrate`, `workspace`, `voice`, `roocode`, `telemetry`, `verifier`, `worker`, `server`) are now broken out into 73 discrete feature rows. `checkpoint`, `telemetry`, most of `verifier`, `server`, and several `worker` features upgrade from round-1 partial to working-untaped on direct-source evidence (real git plumbing, real OTEL exporter, real Raft + cgroup writes, real Gin handlers with integration tests). `roocode` task-delegation/code-gen and `voice/workspace` auto-cleanup remain honest gap (real types, no shipped wiring / incomplete impl).
- **Stub-claim corrections** (see top caveat): three round-2-explorer “stub” reports were verified FALSE by direct read and corrected to done/partial. Logged here so the conductor does not propagate the wrong call.
- **Submodule Real-use=unknown throughout:** each submodule package compiles and has tests, but genuine end-user reachability THROUGH HelixCode (vs the submodule’s own standalone tests) was not confirmed by static inspection — honestly unknown, never green.
- **helix_agent Wired:** yes for the packages HelixCode consumes via the direct `dev.helix.agent` import (llm/ensemble/server/provider/verifier/ streaming surface reachable through the agent dependency); partial for packages reachable only transitively (database/redis/auth/monitoring/etc).
- **panoptic Wired=no:** present as an equal-codebase submodule (CONST-051) but NOT in HelixCode’s Go build graph — it is a Challenge/recording framework consumed out-of-process, not imported.
- Every Real-use=unknown / working-untaped row is a candidate for a recorded scenario; none is video-confirmed (📹 no throughout).

Round-2 deepened feature count: 145 rows (73 internal across 12 packages + 72 submodule across `helix_agent` 19, `security` 11, `helix_specifier` 10, `helix_qa` 14, `panoptic` 10, distributed across the wiring model).

**Sources verified 2026-06-22: `helix_code/internal/*` ,
submodules/
{`helix_agent`,`security`,`helixspecifier`,`helix_qa`,`panoptic`
}/**

REPO-STATE-DERIVED (per §11.4.99 the sources are the cross-referenced repo trees, following the docs/ARCHITECTURE.md precedent — no external service documented). This round-2 deepening re-reads packages round-1 flagged shallow; cross-referenced against the live tree on 2026-06-22: - The deepened rows resolve against real `helix_code/internal/*` package source and the named own-org submodules under `submodules/*` (`helix_agent`, `security`, `helix_qa`, `panoptic`, `helix_specifier`). - The doc already records its own source-read corrections (round-1 stub claims verified false by direct source reads: `workspace` docker stop/remove, `voice` recorder) — those ARE the §11.4.99 negative findings for this slice, captured inline at the head of the file. - No external-service version claims in this doc → no staleness check applies; the evidence is structural (impl reality / wiring / `_test.go` presence).